

$$\begin{aligned}
 &\equiv (p \vee p) \wedge (p \vee \neg q) \quad (\text{1st distributive law}) \\
 c) &(\neg q \rightarrow p) \rightarrow (p \rightarrow \neg q) \equiv (\neg(\neg q) \vee p) \rightarrow (\neg p \vee \neg q) \quad (\text{2 applications of conditional-disjunction equivalence}) \\
 &\equiv (q \vee p) \rightarrow (\neg p \vee \neg q) \quad (\text{Double negation law}) \\
 &\equiv \neg(q \vee p) \vee (\neg p \vee \neg q) \quad (\text{Conditional-disjunction equivalence}) \\
 &\equiv (\neg q \wedge \neg p) \vee (\neg p \vee \neg q) \quad (\text{2nd De Morgan's law}) \\
 &\equiv (\neg p \vee \neg q) \vee (\neg q \wedge \neg p) \quad (\text{1st commutative law}) \\
 &\equiv \neg p \vee (\neg q \vee (\neg q \wedge \neg p)) \quad (\text{1st associative law}) \\
 &\equiv \neg p \vee ((\neg q \wedge \neg p) \vee (\neg q \wedge \neg p)) \quad (\text{1st identity law}) \\
 &\equiv \neg p \vee (\neg q \wedge (\neg p \vee \neg p)) \quad (\text{2nd distributive law}) \\
 &\equiv \neg p \vee (\neg q \wedge (\neg p \vee \neg p)) \quad (\text{and 2nd commutative law}) \\
 &\equiv \neg p \vee (\neg q \wedge (\neg p \vee \neg p)) \quad (\text{1st Commutative law}) \\
 &\equiv \neg p \vee (\neg q \wedge \neg p) \quad (\text{1st Domination law}) \\
 &\equiv \neg p \vee (\neg q \wedge \neg p) \quad (\text{1st identity law}) \\
 &\equiv \neg p \vee \neg q \quad (\text{1st identity law})
 \end{aligned}$$

10) For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not ~~follow~~ involve conditionals.

$$\begin{aligned}
 a) &\neg p \rightarrow \neg q \equiv \neg(\neg p) \vee \neg q \quad (\text{Conditional-disjunction equivalence}) \\
 &\equiv p \vee \neg q \quad (\text{Double-negation law})
 \end{aligned}$$

$$\begin{aligned}
 b) &(p \vee q) \rightarrow \neg p \equiv \neg(p \vee q) \vee \neg p \quad (\text{Conditional-disjunction equivalence}) \\
 &\equiv (\neg p \wedge \neg q) \vee \neg p \quad (\text{2nd De Morgan's law}) \\
 &\equiv (\neg p \wedge \neg q) \vee (\neg p \wedge \neg p) \quad (\text{Identity law 1}) \\
 &\equiv \neg p \wedge (\neg q \vee \neg p) \quad (\text{2nd Distributive Law}) \\
 &\equiv \neg p \wedge \neg p \quad (\text{1st Domination Law}) \\
 &\equiv \neg p \quad (\text{1st Identity law})
 \end{aligned}$$

$$\begin{aligned}
 c) &(p \rightarrow \neg q) \rightarrow (\neg p \rightarrow \neg q) \equiv (\neg p \vee \neg q) \rightarrow (\neg(\neg p) \vee \neg q) \quad (\text{2 applications of conditional-disjunction equivalence}) \\
 &\equiv \neg(\neg p \vee \neg q) \vee (p \vee q) \quad (\text{Double-negation law}) \\
 &\equiv \neg(\neg p \vee \neg q) \vee (p \vee q) \quad (\text{Conditional-disjunction equivalence}) \\
 &\equiv (\neg(\neg p) \wedge \neg(\neg q)) \vee (p \vee q) \quad (\text{2nd De Morgan's Law}) \\
 &\equiv (p \wedge q) \vee (p \vee q) \quad (\text{2 applications of Double-negation Law})
 \end{aligned}$$